MSML612 Project Proposal (Group-9)

Identity-Preserving Story Visualization via Diffusion Models
with Decoupled Identity Conditioning

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1. Problem Statement

Story visualization is the task of generating a coherent sequence of images from a series of textual captions that form a narrative. A critical requirement is visual consistency: characters must maintain their appearance: facial features, clothing, body shape, and color palette: across all generated frames. Without this, the image sequence fails to convey a coherent story.

Current state-of-the-art methods such as StoryDiffusion [1] achieve improved consistency through shared self-attention mechanisms, but still exhibit noticeable identity drift, particularly in longer sequences (6+ frames) and multi-character scenes. We propose to address this by augmenting existing story visualization pipelines with an explicit identity conditioning mechanism. Specifically, we will integrate identity-preserving features, extracted using adapters such as IP-Adapter [4], into the diffusion process via decoupled cross-attention, aiming for measurable improvement in character consistency without sacrificing image quality or text-image alignment.

2. Relevant Published Work

StoryDiffusion (Zhou et al., NeurIPS 2024) [1] introduces Consistent Self-Attention, enabling character-consistent generation over long-range sequences. It is compatible with SD 1.5 and SDXL backbones and serves as our primary baseline.

AR-LDM (Pan et al., WACV 2024) [2] is the first work to leverage latent diffusion models for story visualization, using an autoregressive approach conditioned on previous captions and generated images. It achieves strong FID scores on PororoSV, FlintstonesSV, and VIST, and serves as our secondary baseline.

Make-A-Story (Rahman et al., CVPR 2023) [3] proposes a visual memory module that captures actor and background context across frames through sentence-conditioned soft attention. Its memory-based approach inspires our strategy of explicitly injecting character identity features.

IP-Adapter (Ye et al., 2023) [4] introduces a lightweight adapter (~22M parameters) with a decoupled cross-attention mechanism that separates image features from text features. This allows image prompts to work alongside text prompts without interference. We plan to adapt this mechanism to inject character identity features into the story generation pipeline.

TemporalStory (Chen et al., 2024) [5] uses Spatial-Temporal attention instead of autoregressive approaches, with a text adapter and StoryFlow adapter for cross-sentence context. Its evaluation framework on PororoSV and FlintstonesSV provides benchmarks for comparison.

3. Scope and Approach

3.1 Proposed Architecture

We augment the StoryDiffusion pipeline with an identity preservation module consisting of three components:

Identity Encoder. Given a reference image of each character (from the first generated frame or provided as input), we extract identity features using a pretrained encoder. We will compare: (a) IP-Adapter's CLIP image encoder for general visual features, (b) DINOv2 for semantically rich appearance matching, and (c) InsightFace for face-specific embeddings.

Decoupled Cross-Attention Injection. Following the IP-Adapter design, we add a parallel cross-attention layer to the U-Net that attends to identity features separately from text cross-attention. This ensures character appearance information does not interfere with the text prompt's control over scene composition. The identity attention scale is adjustable to balance identity preservation against text fidelity.

Per-Character Conditioning. For multi-character scenes, we investigate per-character identity injection using spatial masks or regional cross-attention, allowing different image regions to attend to different character embeddings.

3.2 Datasets

PororoSV (primary): 10,191 training / 2,334 validation / 2,208 test stories, each with 5 consecutive captioned frames and 9 recurring characters. The most widely used benchmark in story visualization. **FlintstonesSV** (secondary): used for cross-dataset generalization evaluation. Both are publicly available with standardized splits.

3.3 Evaluation

We adopt the standard evaluation protocol from prior work [6], supplemented with identity-specific metrics:

- **FID** - distributional distance between generated and real images (image quality).
- **Character F1** - proportion of correctly generated characters per frame, using a fine-tuned Inception-v3 classifier.
- **Frame Accuracy** - percentage of frames where all specified characters are present (stricter than Char F1).

- **Cross-Frame CLIP/DINOv2 Similarity (proposed)** - cosine similarity of character crops across frames, directly measuring identity consistency.

References

[1] Y. Zhou et al. "StoryDiffusion: Consistent Self-Attention for Long-Range Image and Video Generation." NeurIPS 2024.

[2] X. Pan et al. "Synthesizing Coherent Story with Auto-Regressive Latent Diffusion Models." WACV 2024.

[3] T. Rahman et al. "Make-A-Story: Visual Memory Conditioned Consistent Story Generation." CVPR 2023.

[4] H. Ye et al. "IP-Adapter: Text Compatible Image Prompt Adapter for Text-to-Image Diffusion Models." arXiv:2308.06721, 2023.

[5] Z. Chen et al. "TemporalStory: Enhancing Consistency in Story Visualization using Spatial-Temporal Attention." arXiv:2407.09774, 2024.

[6] A. Maharana and M. Bansal. "Integrating Visuospatial, Linguistic and Commonsense Structure into Story Visualization." EMNLP 2021.